

Data Visualization. Exploratory Data Analysis

Seeing what's inside our data and allowing others to see



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SoftUni



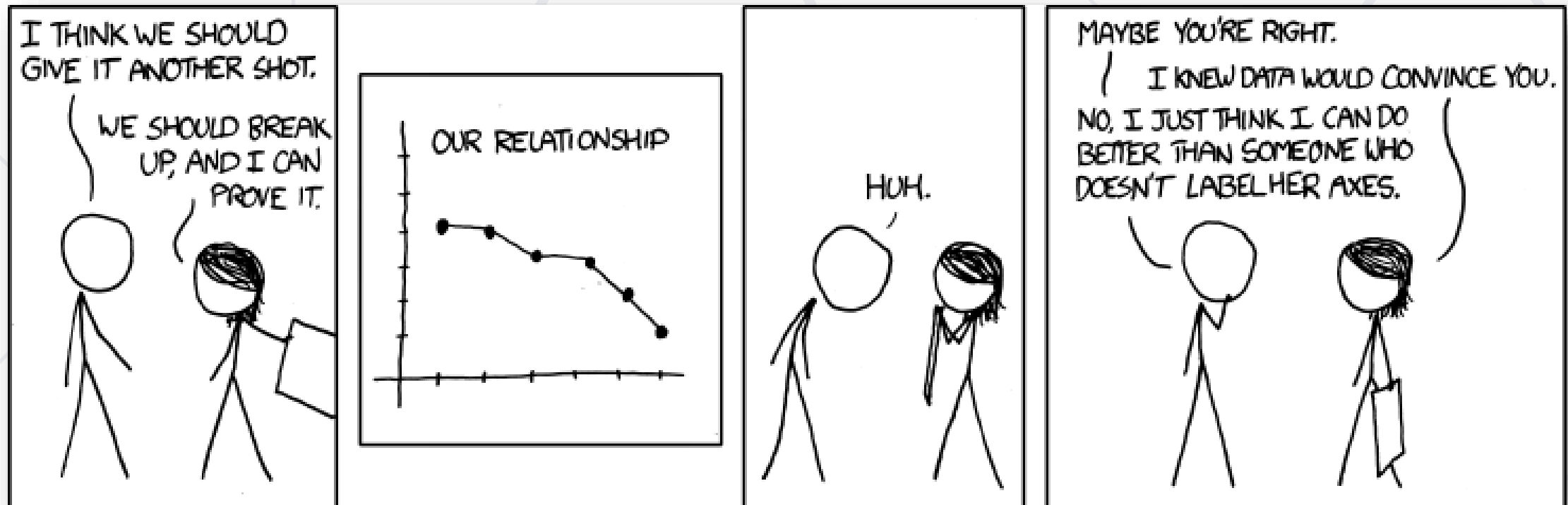
Software University

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1. Main Concepts and Rules
2. Creating Simple Plots
3. Real-Life Examples: Good and Bad
4. Customizing Plots
5. Exploratory Data Analysis
 - Basic Guidelines
 - EDA as Part of the Data Science Process



Be Careful...

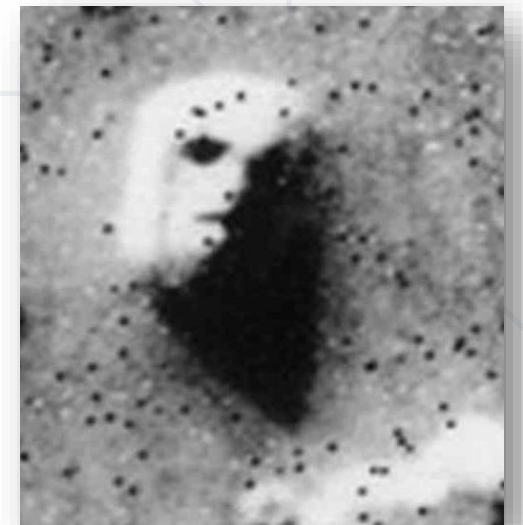
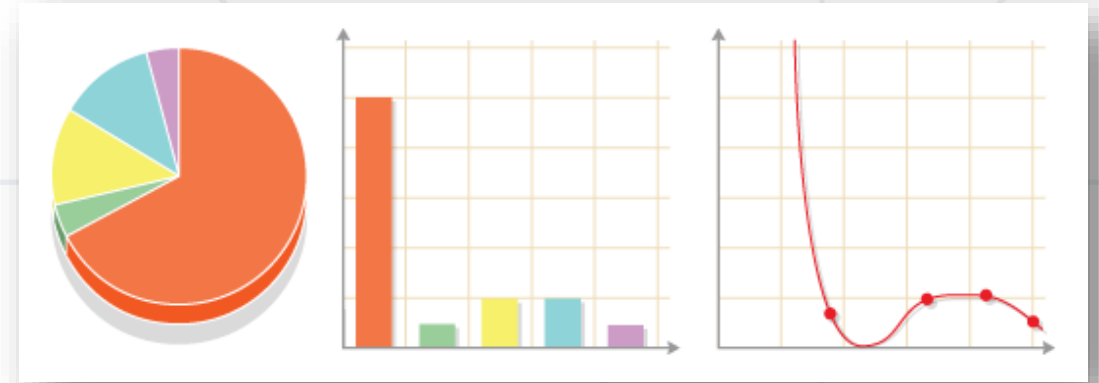




Main Concepts in Data Visualization

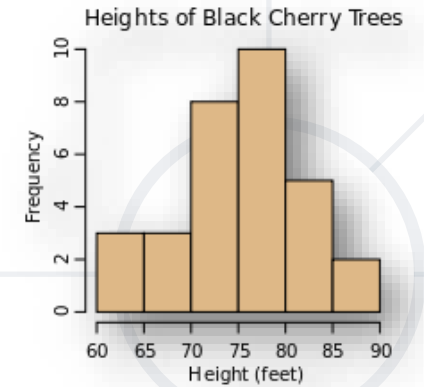
How to Tell the Right Story

- We're amazingly good at spotting patterns
 - Trends over time, correlations, comparisons, ranges, etc.
- Visualizing data helps us understand the information better
- By plotting different views on the data we can
 - Help ourselves explore and understand the data
 - Convey the stories in data to others
- Note that we're too good at spotting patterns
 - We can find patterns where they don't exist



Knowing What (and When) to Plot

- Many types of graphs, each with its own purpose
 - **Histograms** – show distributions
 - **Boxplots** – show the range and skewness of values
 - **Bar charts** – show how different categories compare
 - **Line plots** – show how one (dependent) variable varies with respect to an independent variable (e.g. over time)
 - **Pie charts** :(– show relative sizes between parts of a whole
 - Don't forget that we can also display **single numbers** when they provide sufficient information



Perceived Value for Money



Knowing What (and When) to Plot

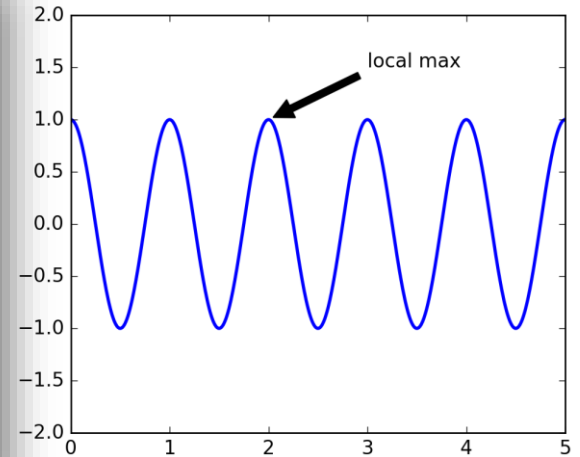
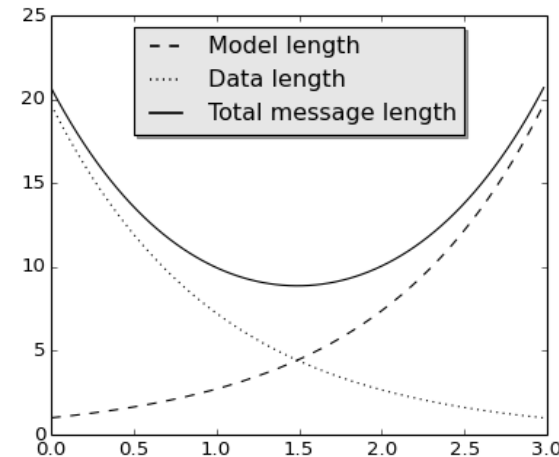
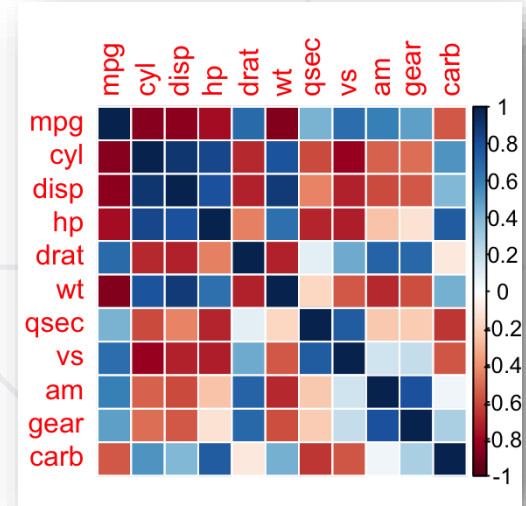
- Many more types of graphics depending on the context
- Choosing the right plot is a matter of **intuition**
 - The goal is to **present the message clearly**
 - I.e., "**tell the right story**"
- Two main kinds of visualizations
 - For scientific analysis and work – stricter rules
 - For exploratory analysis / quick references
 - For presenting results to non-specialists – we can be creative but we have to keep our message in mind
 - The results may be printed or viewed as a **dashboard**
- How many dimensions?
 - Each plot has two spatial dimensions, but we can add more using color, size, even animation

Knowing What (and When) to Plot

- To choose a plot type, think about
 - Numerical or categorical variables
 - Structure – spatial, temporal, etc.
 - Clustering
 - Relative size
- How can you know what all of these are?
 - Perform an **exploratory data analysis** first
 - "Play around" with the data
 - Check different measures (such as means, standard deviations, ranges, etc.), plot different charts
 - Explore the distributions and relations of variables
 - **Document** the **exploration process** and your **findings** to remember them later

Basic Rules

- Choose the appropriate chart type
 - If you can (and want), compare different types of charts
- Make your plot big enough to fit the plotting area
- When it's not obvious, add a title and a legend
- **Label the axes!**
- Optionally, point at interesting data
- Use marker size and color to convey information
- Don't strain the reader!





Plotting Basics

Starting out Easy

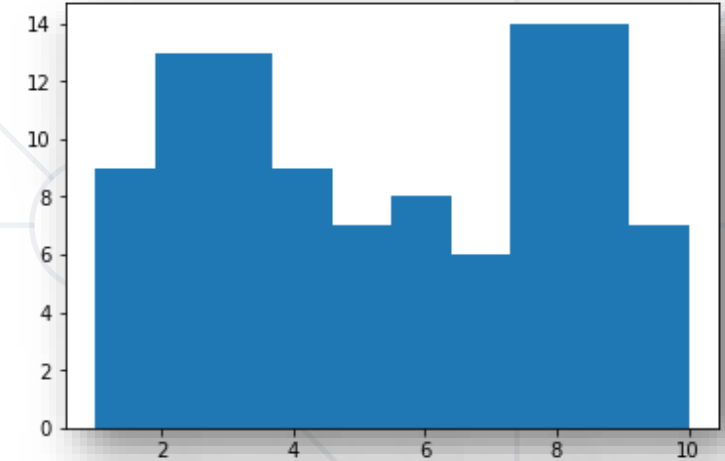
- Quite easy to start plotting
- Very powerful
- There are many ways to do the same thing
- The documentation and examples are really good
 - We'll often end up consulting them, or the community
 - There are many options and it's difficult to remember them all
- Importing the library

```
import matplotlib.pyplot as plt
```
- In Jupyter notebook, write the magic string `%matplotlib inline` in the first cell before importing
 - This will make plots appear as images in the notebook

■ Histogram

- Shows the distribution of one variable

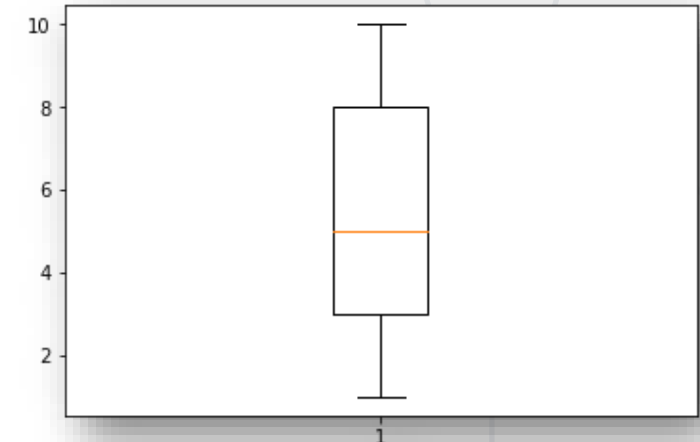
```
import numpy as np
values = np.random.randint(1, 11, 100)
plt.hist(values)
plt.show()
```



■ Boxplot

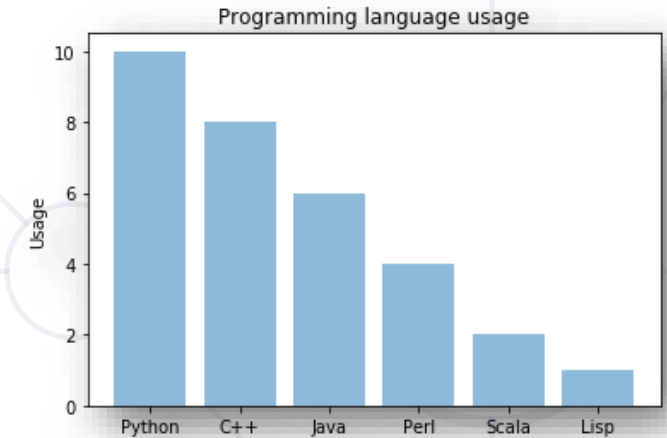
- Another way to show the distribution of one variable
- May also be used to compare many distributions
- How to read a boxplot

```
plt.boxplot(values)
plt.show()
```



Creating Simple Plots

- Bar chart
 - Shows how one numeric value compares among different categories
 - Two variables: one categorical, one numerical
 - A little bit more difficult to plot

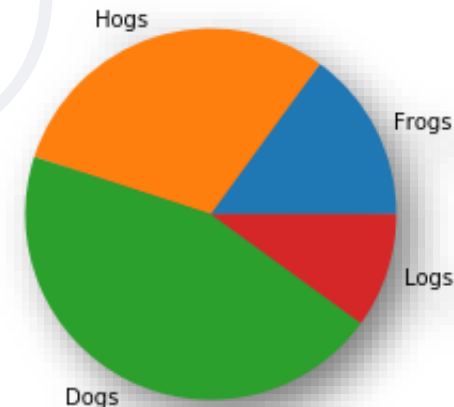


- **Although they look similar, histograms and bar charts are different!**

- Pie chart

- Shows the relation of each part to the whole

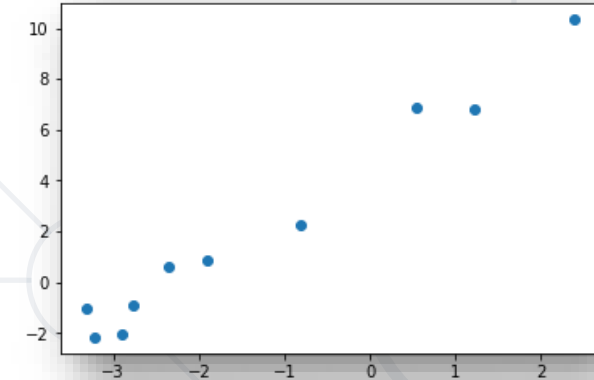
```
sizes = [15, 30, 45, 10]
plt.pie(sizes, labels = ["Frogs",
                        "Hogs", "Dogs", "Logs"])
# Make the plot look circular
plt.gca().set_aspect("equal")
```



Creating Simple Plots

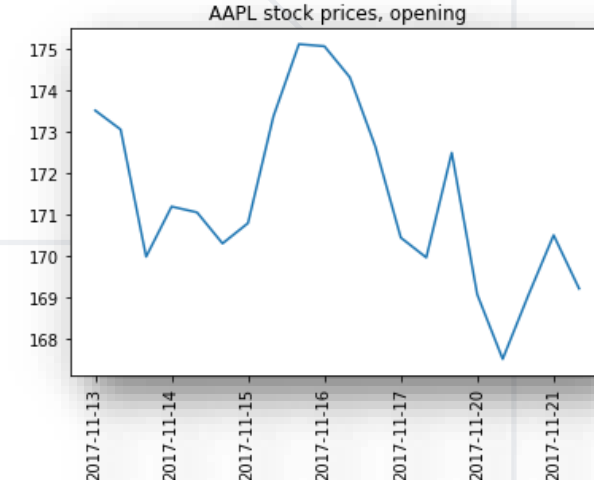
- Scatterplot (or scatter plot)
 - Shows how two variables compare
 - Can be used for displaying trends or correlations

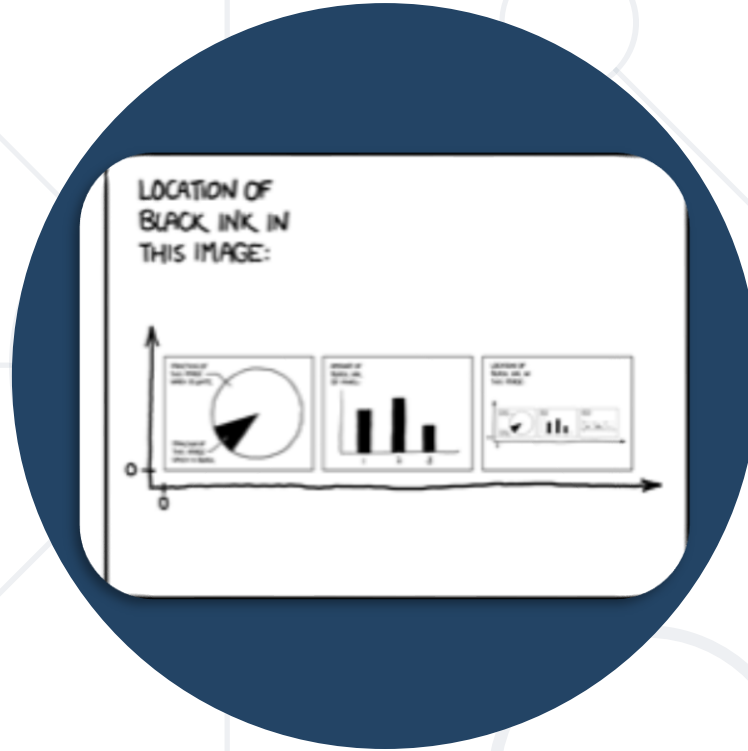
```
x = [-2.35, 1.22, -3.32, 2.39, -2.77,  
      -3.21, 0.55, -0.81, -2.89, -1.9]  
y = [0.58, 6.79, -1.01, 10.32, -0.9,  
      -2.16, 6.87, 2.22, -2.05, 0.86]  
plt.scatter(x, y)  
plt.show()
```



- Line chart
 - Similar to scatterplot
 - Useful to show dependencies of two variables
 - If the horizontal axis is time – evolution

```
dates = ...  
open_prices = ...  
plt.plot(dates, open_prices)  
plt.xticks(dates[::3], rotation = "vertical")  
plt.title("AAPL stock prices, opening")  
plt.show()
```

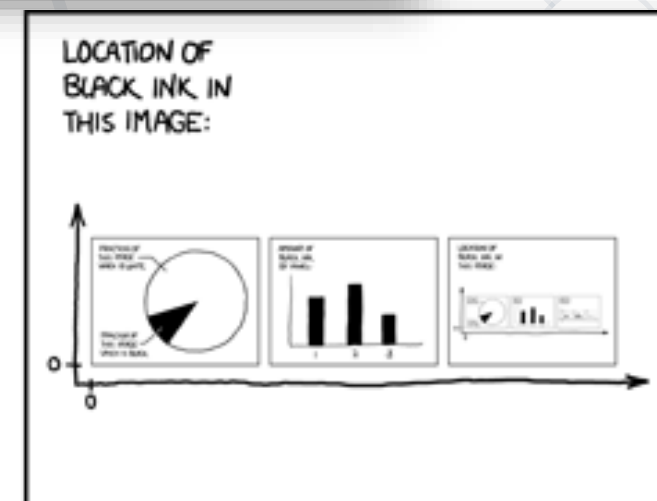
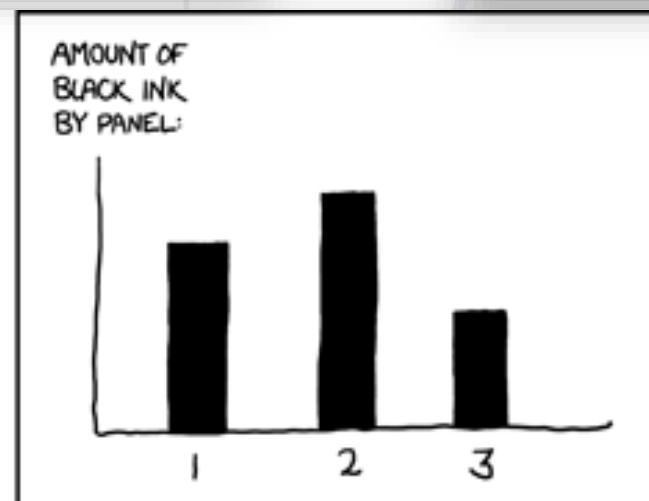
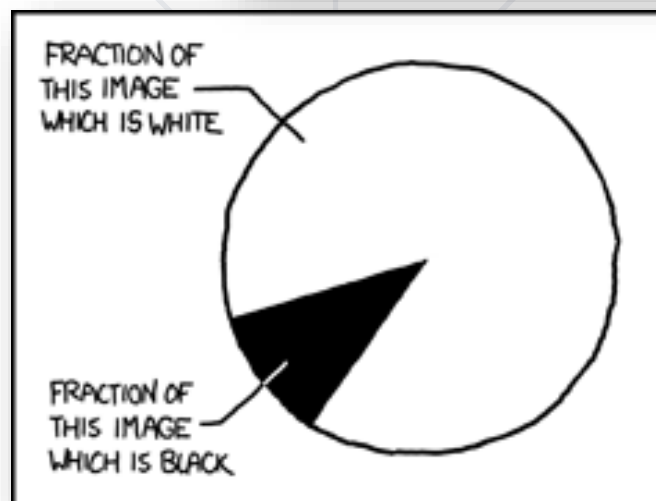
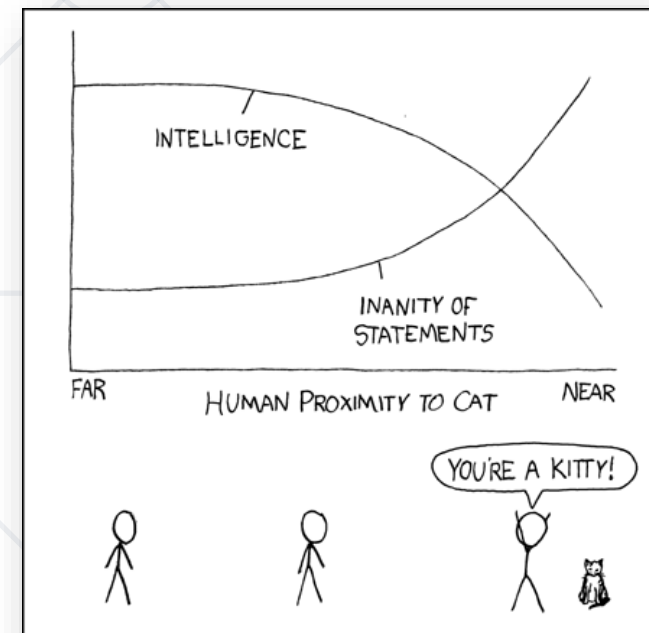
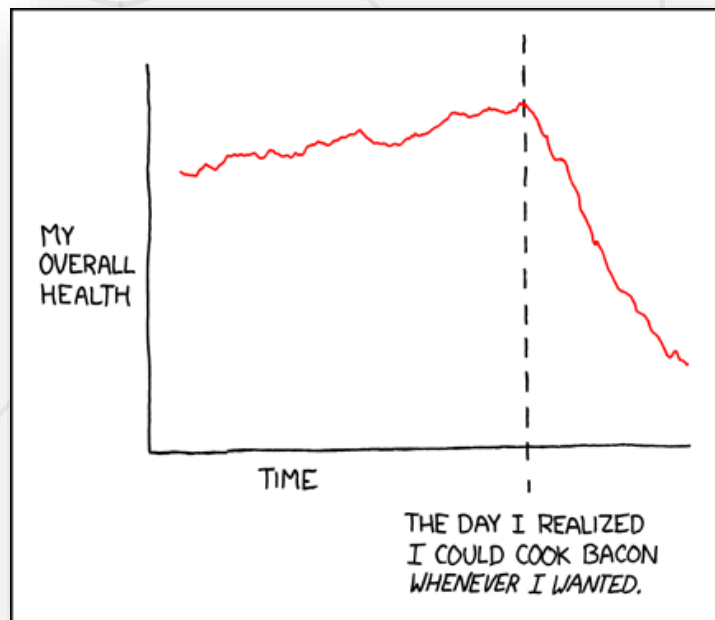




Data Visualization Examples

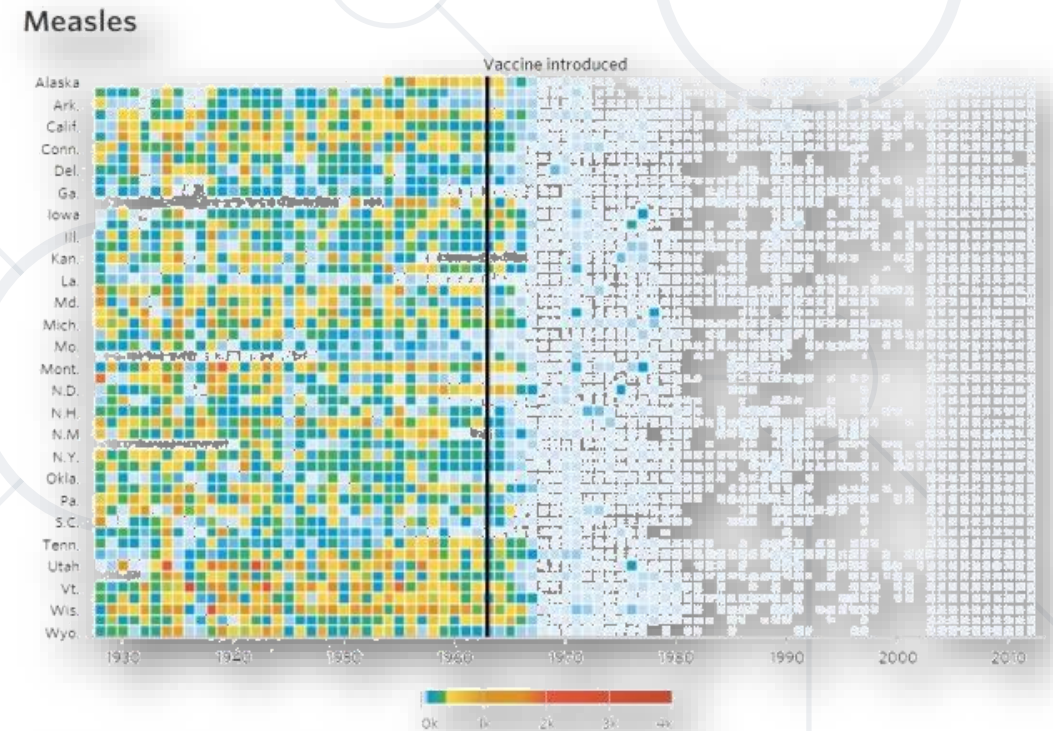
The Good, the Bad, the Ugly and the WTF

Some Examples



Infectious Diseases and Vaccines ✓

- Once again, a heatmap is used to convey disease spread
 - The legend has numbers
 - The full chart is interactive(provides all numbers)
- Makes use of temporal data
- Labels an interesting point
 - Allows us to compare "before" / "after"
- Conveys a clear message
 - Vaccines almost eradicated diseases



Note: CDC data from 2003-2012 comes from its Summary of Notifiable Diseases, which publishes yearly rather than weekly and counts confirmed cases as opposed to provisional ones.

Gay Marriage Acceptance ✓

- Good use of spatial structure (heatmaps)
- Overall message is clear
 - All states are in the "more" category
 - Some states are more accepting
- May additionally display numbers on the scale or on the map to give a quantitative view
 - This is an editorial, not a scientific plot so it's acceptable

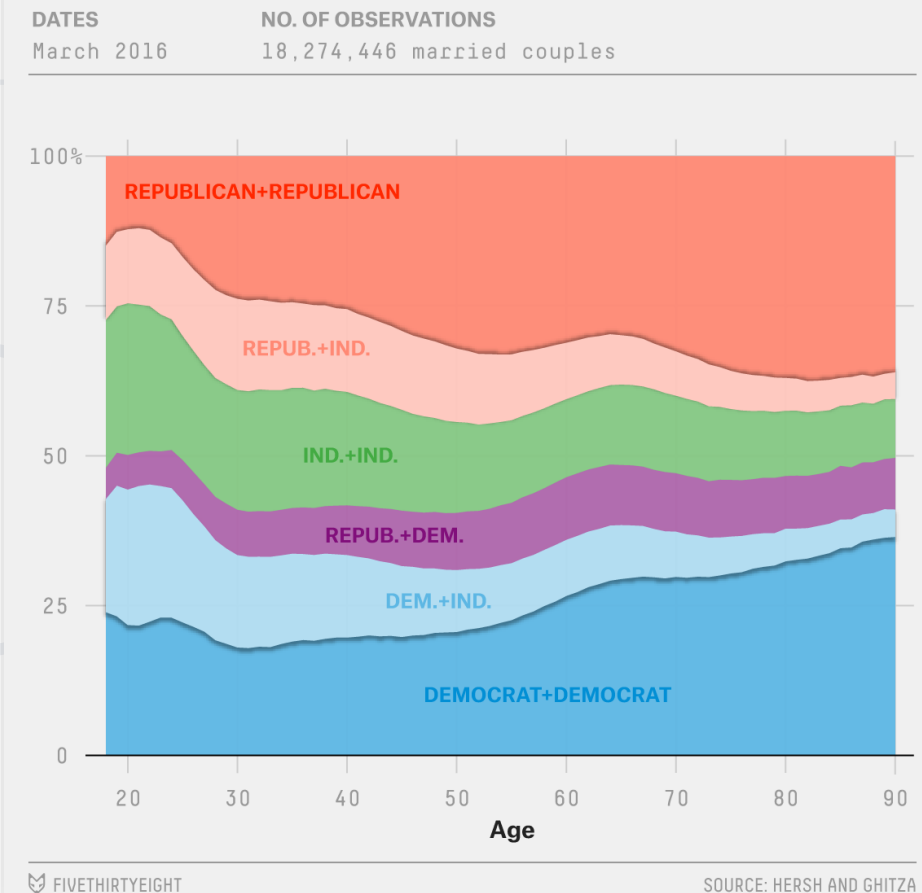


Political Views of Couples ✓

- Source: [FiveThirtyEight](#)
- Uses area plots to display relations
 - Allows comparison of different distributions for different ages
- Uses a clear color map
- Uses labels to make comparisons easier

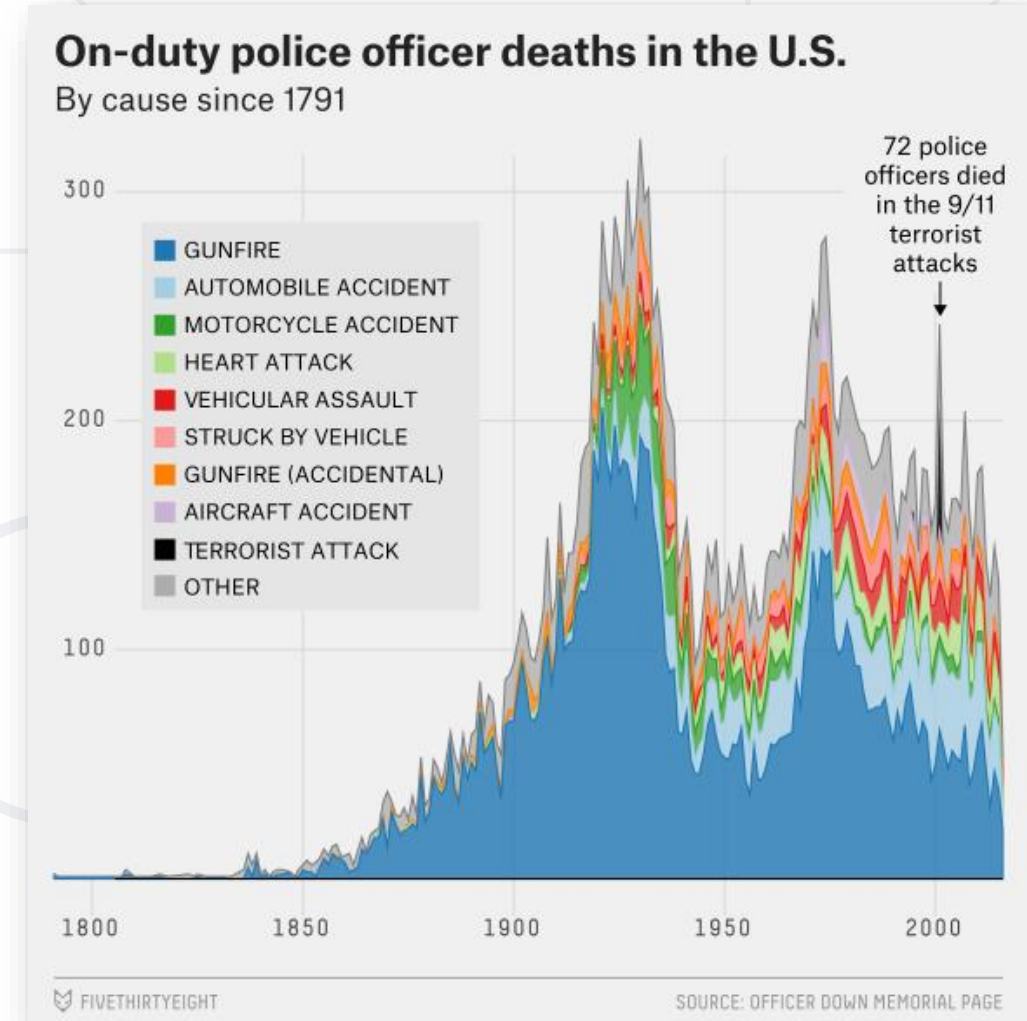
Older couples are more partisan

Share of married couples with different political compositions, by average age



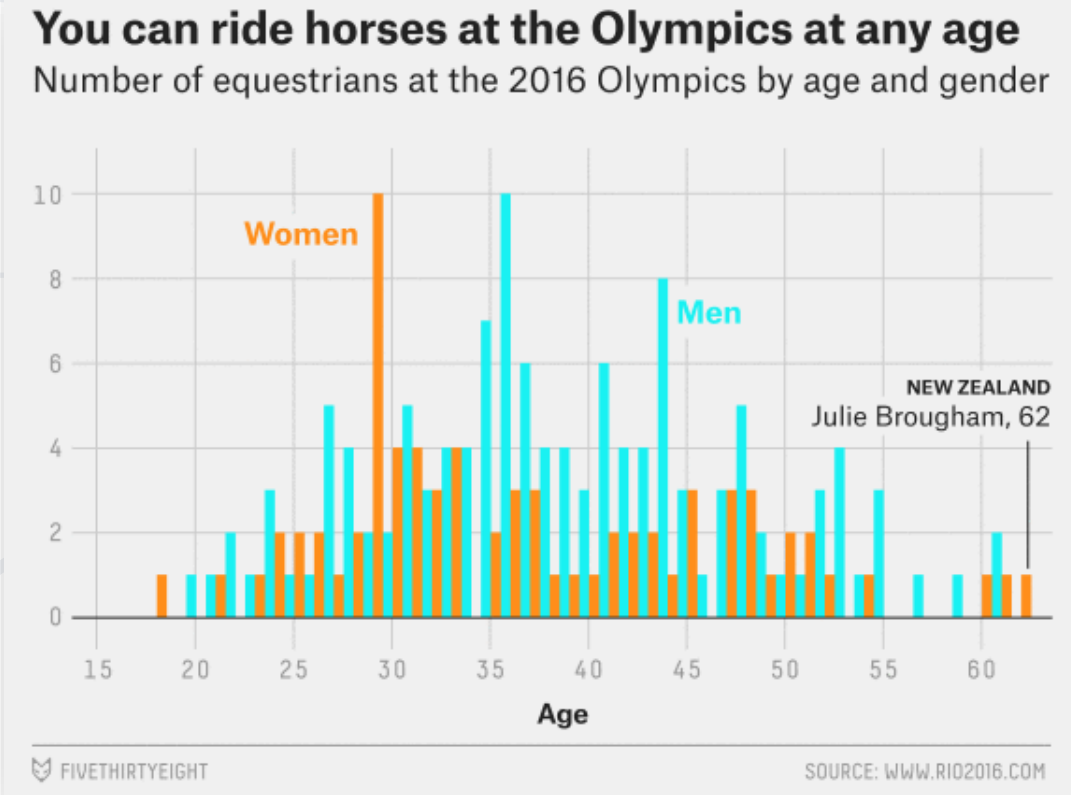
On-Duty Officer Deaths in the US ✓

- Source: [FiveThirtyEight](#)
- Uses a stacked area chart
 - Total area – **overall** tendency
 - Colored parts – tendency **for different causes**
 - Allows to inspect both
- Labels an interesting point



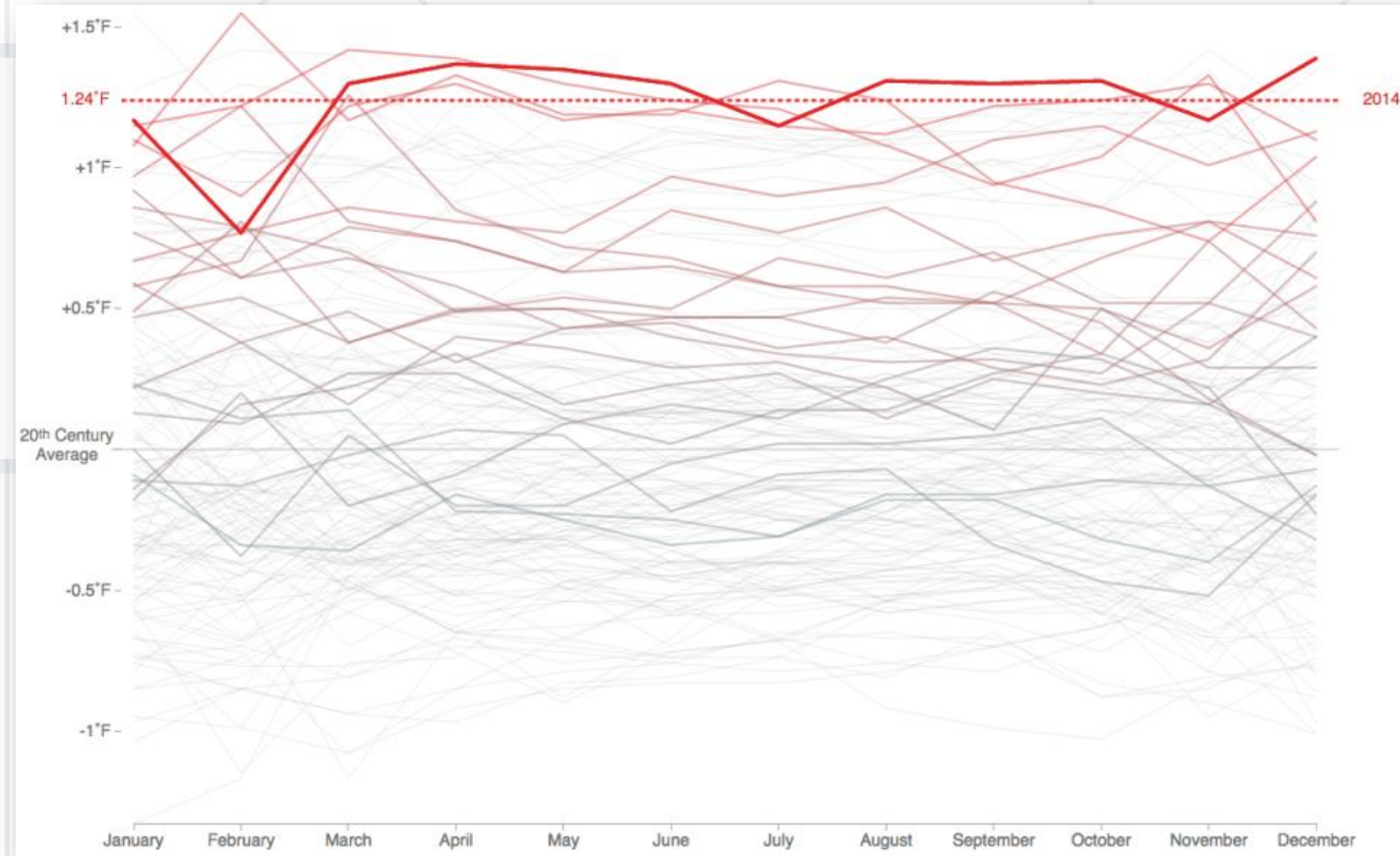
Horse Riders by Age and Gender ✓

- Source: *FiveThirtyEight*
- Histograms are relatively rare in non-scientific visuals
- Shows the two distributions clearly
- Conveys the message
 - The distributions are nearly identical
- Uses labels for outliers



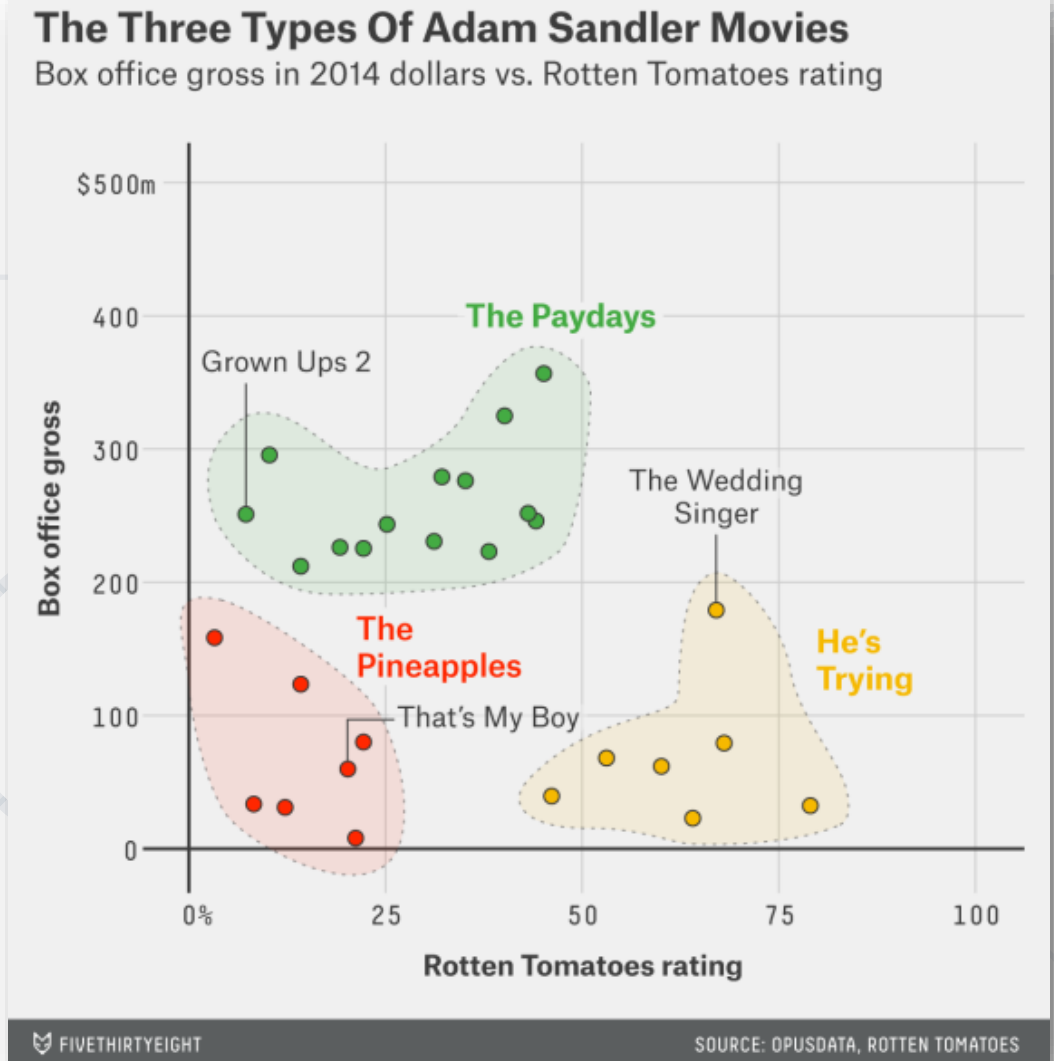
2014: The Hottest Year on Record ✓

- Source: [FlowingData](#)
- Presents temporal data in a classic way
- Uses color to show rising temperature
- Uses a thicker line to make it stand out in an otherwise very busy line plot



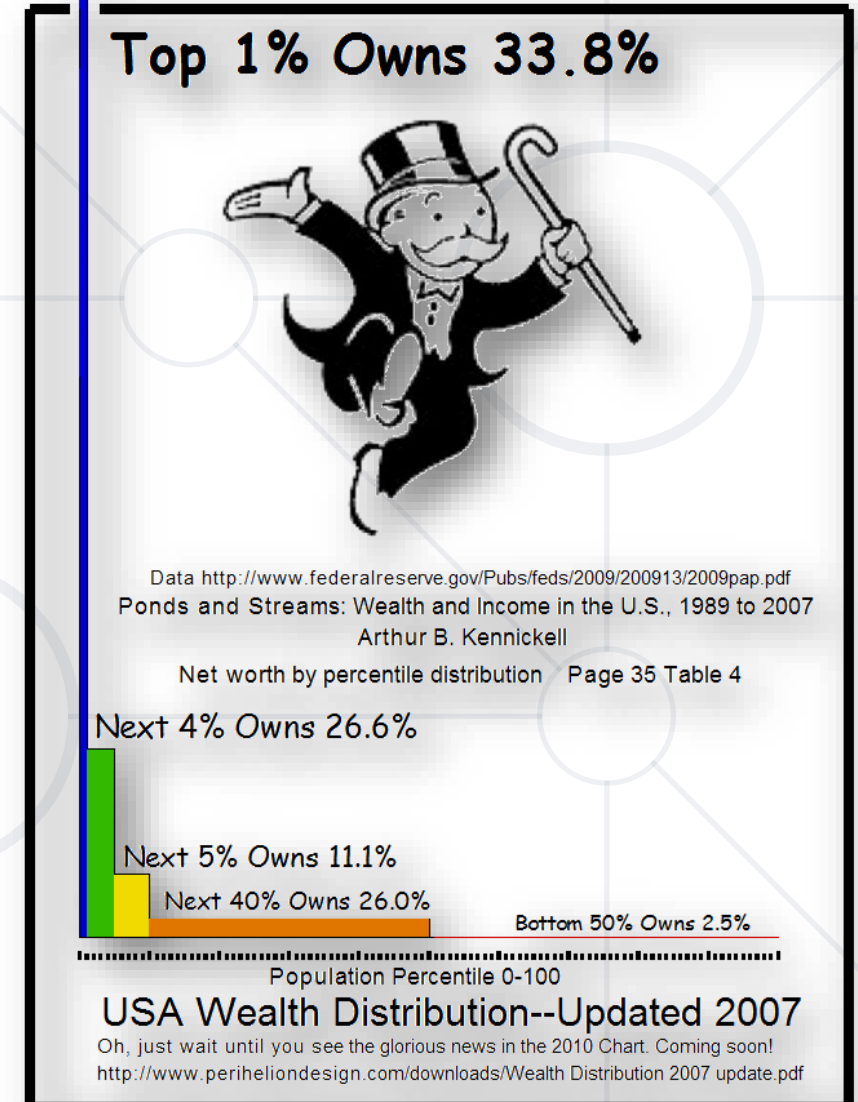
Types of Adam Sandler Movies ✓

- Source: [FiveThirtyEight](#)
- Presents a scatterplot of rating and profits
- Shows and labels clusters clearly
 - Uses different colors
- Conveys a clear message



USA Wealth Distribution X

- Source: WTFViz via Gizmodo
- Highly skewed and disproportional elements
 - It wants to convey the message of disproportionality but there are better ways (e. g. "cutting" the y-axis and displaying y-axis labels)
- Comic Sans(?!), useless image and useless text right in the middle of the chart



Too Much Pie X

- Source: WTFViz via DesignRoast
- The parts of the pie add up to 188%
 - They're meant to be viewed on their own, e.g. there might be combinations of factors
- A pie chart is highly **not recommended** in this case
- Other than that, it shows good labels and a nice color scheme



Too Much Pie, Part 2 **X**

- Source: WTFViz (New Yorker)
- Once again, the pie chart makes no sense
- The values aren't related at all
- Why is there a world map?



Too Much Pie, Part 3 ✗

- Source: Email from GoDaddy, WTFViz
- The numbers aren't related to the color of the rings at all
- Maybe just a programming mistake
 - Still, be careful and check your work



of domain owners have put moderate to high consideration to how much their domains are worth



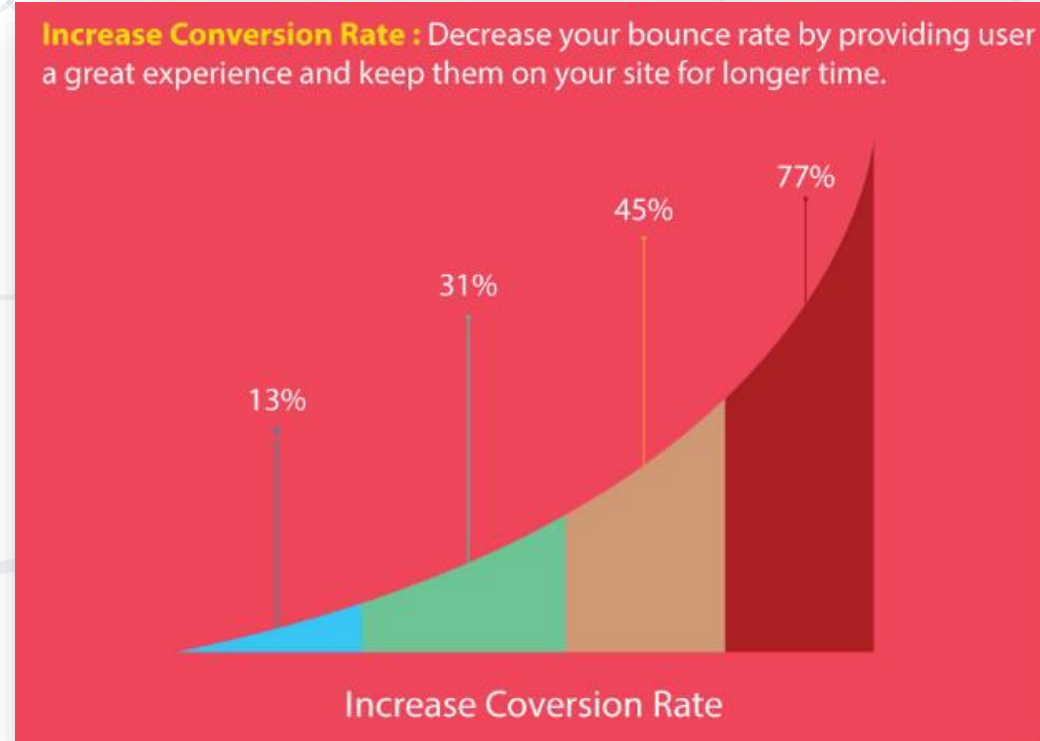
of respondents have bought and sold domain names for a profit



of small business owners want to make time to enhance or update their online presence

No Axes **X**

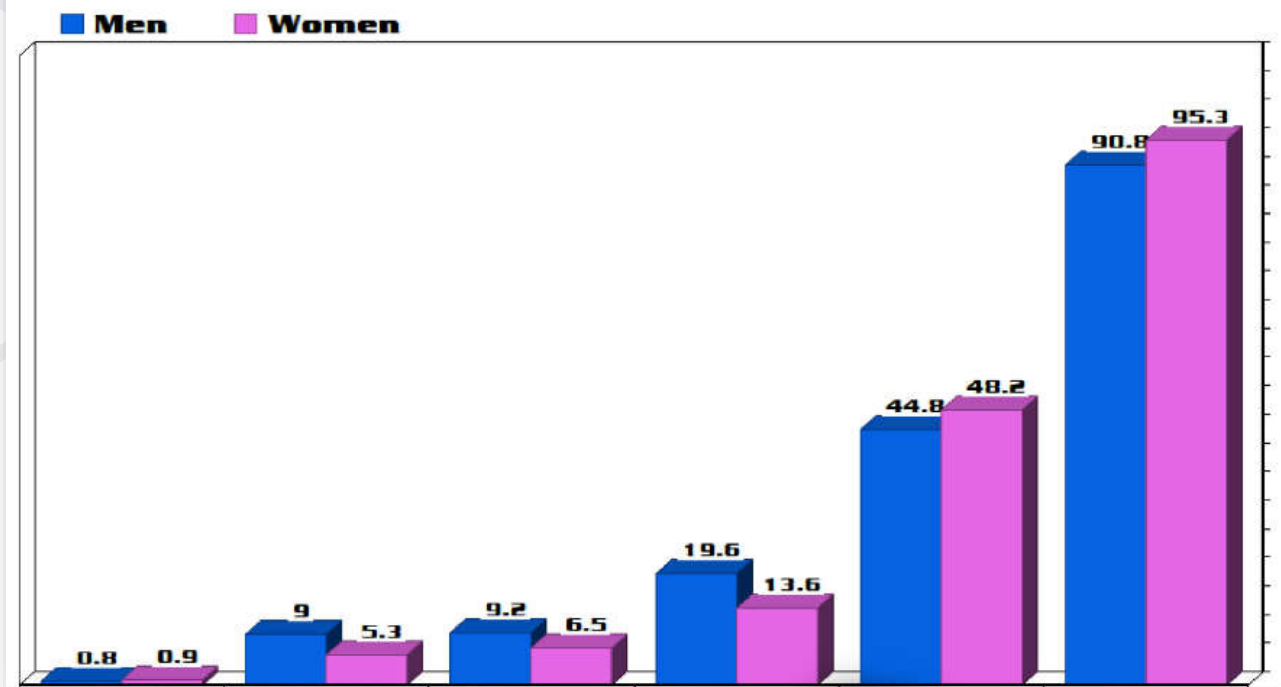
- Source: WTFViz via DesignRoast
- I don't know (and can't see) the real purpose
- No axis labels and no numbers (the bottom label is not the x-axis)
- Distracting background
 - Do you remember "Don't strain the reader"?



No Axes, Part 2 X

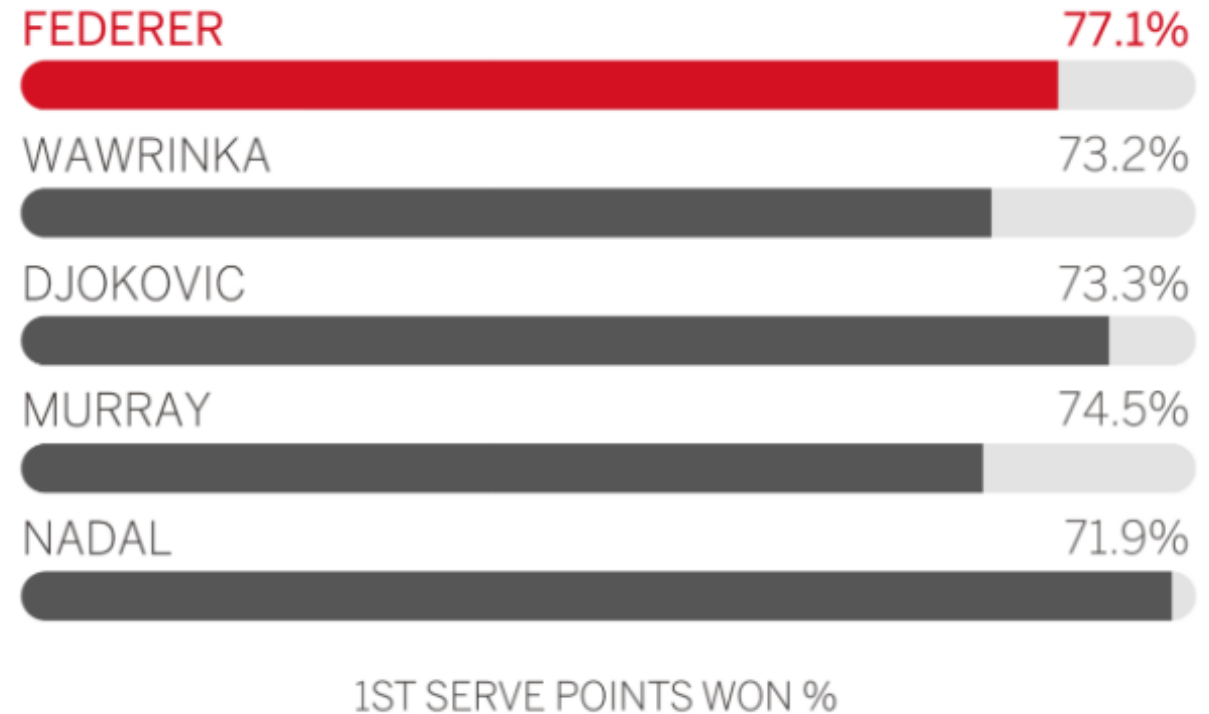
- Source: WTFViz via DesignRoast
- The categories are gone
 - The image is not trimmed, this is the entire chart
 - Are those different days, different products or something else?
 - Also, 3D doesn't give additional information
- Also, the design is kind of lame

Shipping Service Usage During Holidays (%)



Wrong Scales **X**

- Source: WTFViz via DesignRoast
- How come 71,9% is
 - Further than 77,1%
 - Close to full (looks like 95-98%)



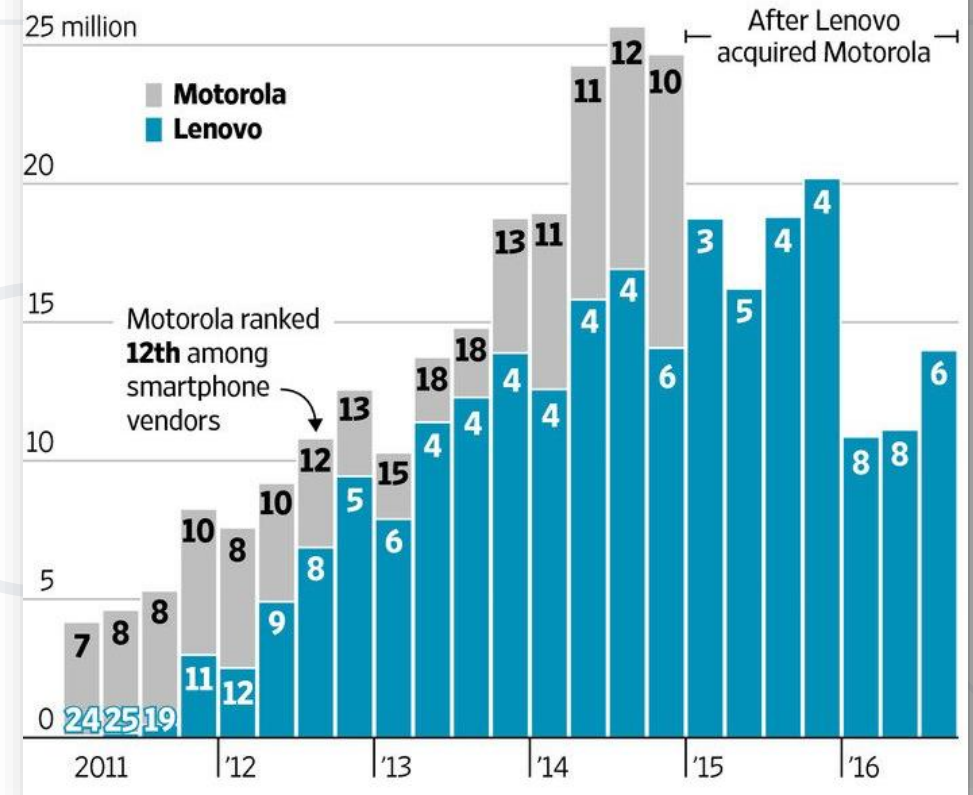
Double Scales that Make No Sense X

- Source: WTFViz
- The y-axis scales and the numbers on the bars represent different things
 - They're also on different scales (blue 11 is less than gray 10 but blue 4 seems very close to gray 18)
- Impossible to read and understand without additional explanation

Smartphone Hang-ups

China's Lenovo Group has struggled to integrate the smartphone business of Motorola, which it acquired in October 2014, part of a surge in Chinese acquisitions of foreign companies.

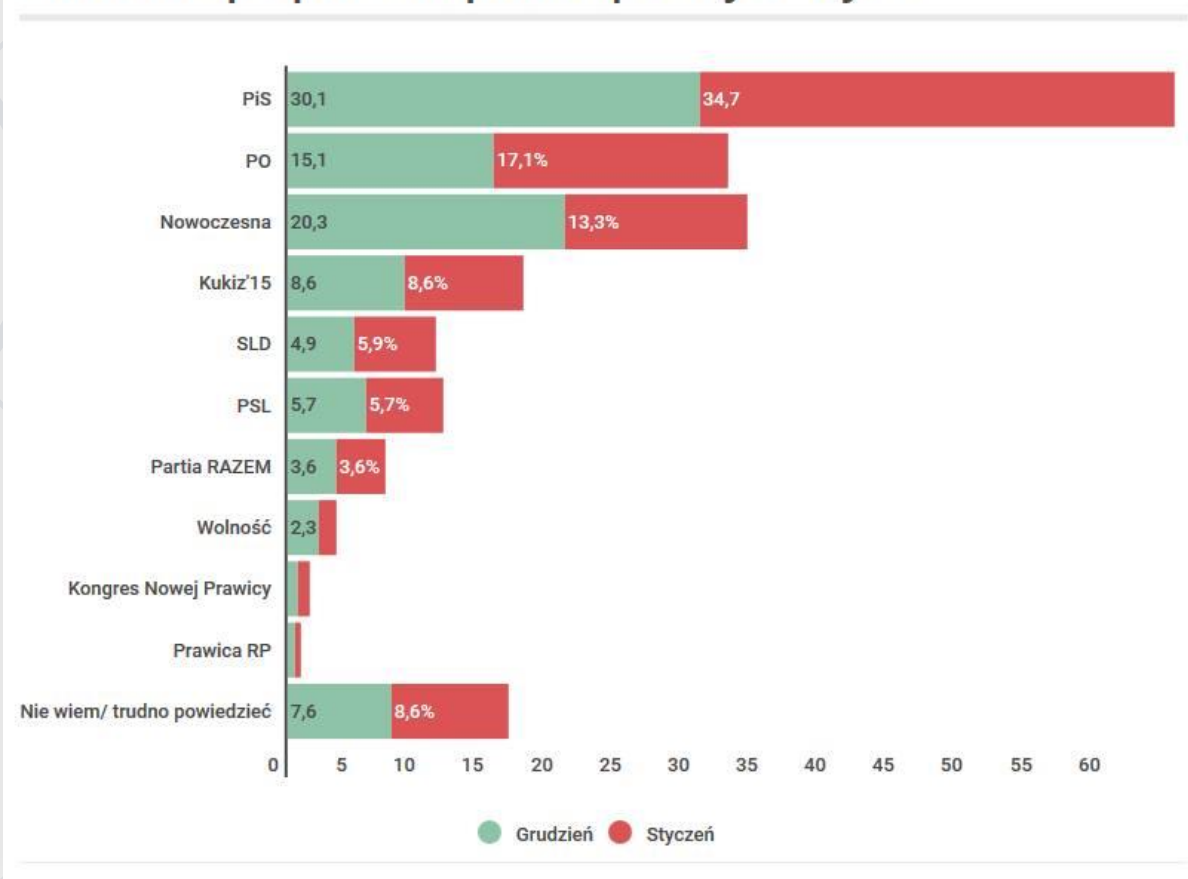
World-wide smartphone shipments and vendor ranking



Wrong Data X

- Source: WTFViz
- The chart looks OK
- Political party affirmation for December (blue) and January (red)
 - Why would one sum percentages like these?
Makes no sense

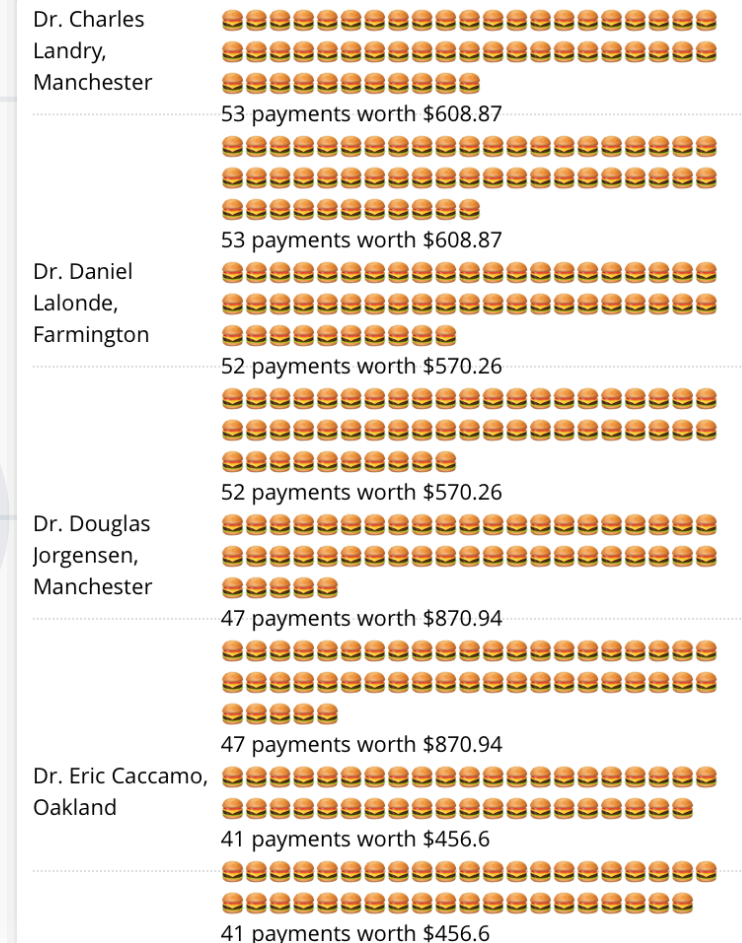
Sondaż poparcia partii politycznych



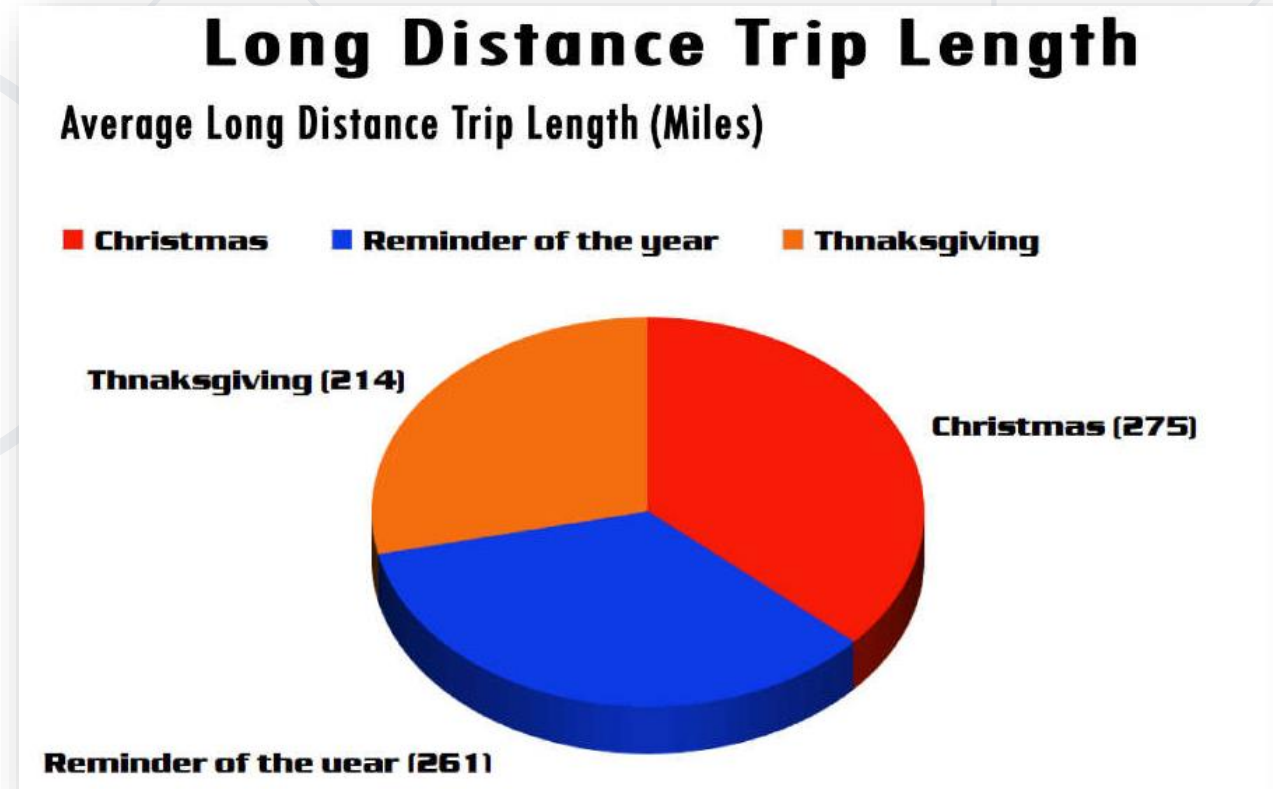
Wrong Data X

- Source: WTFViz
- Those burgers make data extremely difficult to compare
 - The numbers don't help very much

Top 10 doctors with the most “food and beverage” payments from opioid manufacturers, Aug. 2013 – Dec. 2015:

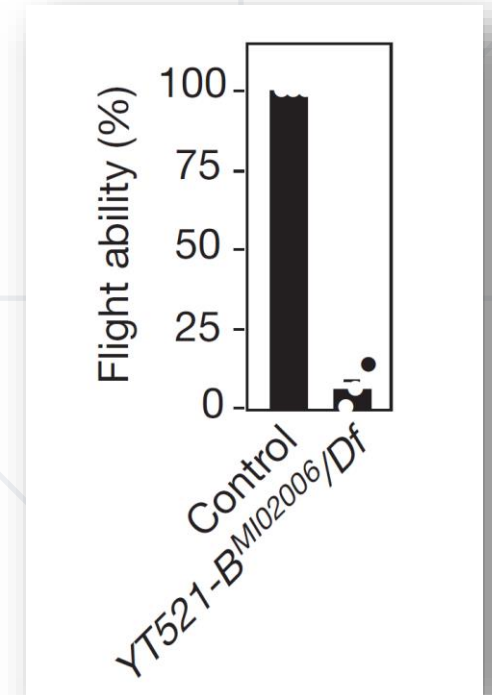


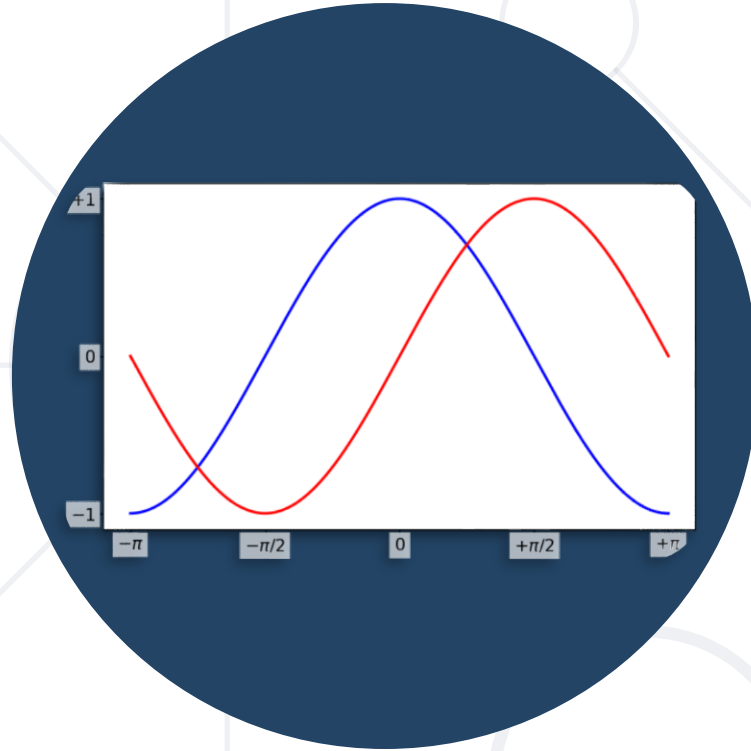
- Source: WTFViz
- Spelling errors
 - Also, unreadable font
- The pie chart conveys no information at all
 - Better – use a bar chart



Bar Chart Mistakes X

- Source: WTFViz
 - Original: Nature :(
- Bubbles make the values extremely difficult to compare
 - Where does the right bar end?
 - Where does even the left bar end?
- Below: Why are the bars warped?





Customizing Plots

Making Things Beautiful

Applying Styles

- Matplotlib has many default styles

```
print(plt.style.available)
```

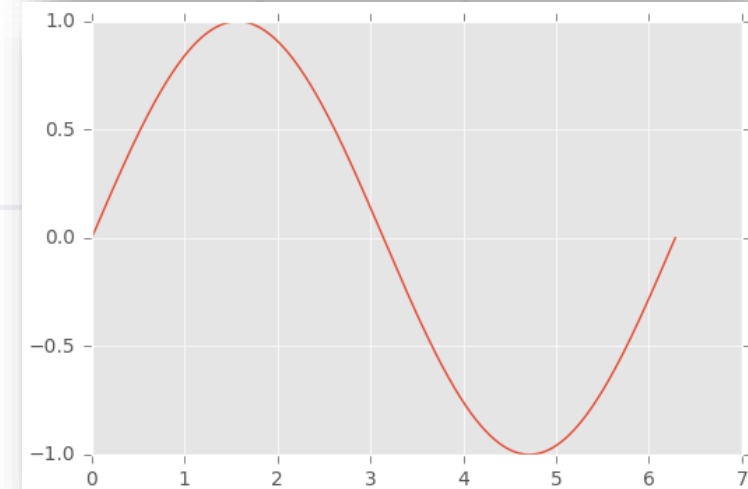
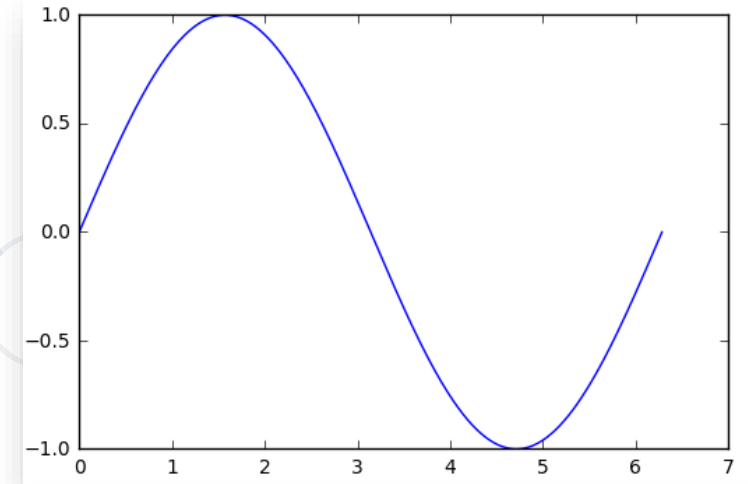
- Using a different style

```
plt.style.use("ggplot")
```

- Reverting to the default style

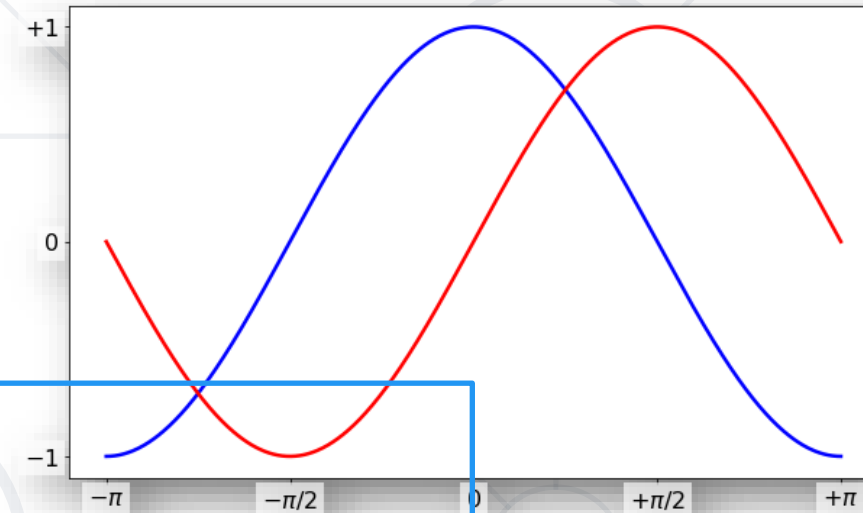
```
plt.style.use("default")
```

- In Jupyter notebook, `%matplotlib inline` uses its own styles
- Example: Draw a simple scatterplot, histogram and / or line chart in all different styles



- Every call to `plt.hist()`, `plt.plot()`, `plt.boxplot()`, etc., accepts many arguments
 - Colors, markers (type, size)
 - Axis limits and locations, axis labels
 - Data labels and additional text
 - Legend location and appearance

```
cos_x = ... # [-pi; pi]
sin_x = ...
plt.figure(figsize = (10, 6))
plt.plot(x, cos_x, color = "blue", linewidth = 2.5, linestyle = "-")
plt.plot(x, sin_x, color = "red", linewidth = 2.5, linestyle = "-")
# Tick marks and labels
plt.xticks([-np.pi, -np.pi / 2, 0, np.pi / 2, np.pi],
           [r"$-\pi$", r"$-\pi/2$", r"$0$", r"$+\pi/2$", r"$+\pi$"])
plt.yticks([-1, 0, 1], [r"$-1$", r"$0$", r"$+1$"])
for label in ax.get_xticklabels() + ax.get_yticklabels():
    label.set_fontsize(16)
    label.set_bbox({facecolor: "white", edgecolor: "None", alpha: 0.65})
```

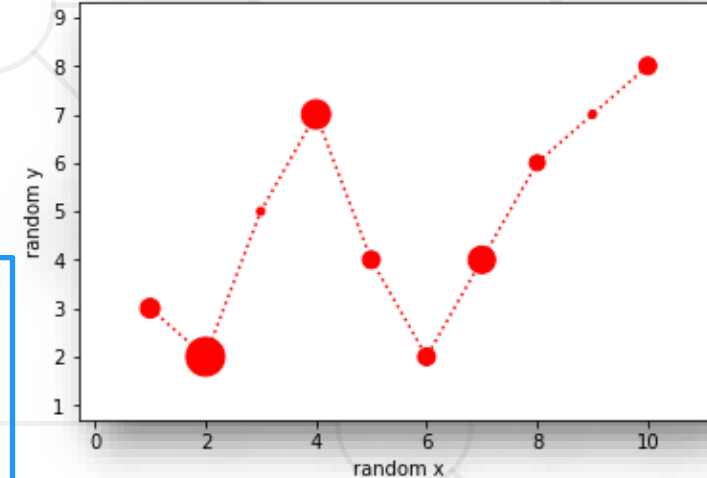


Example: Create a Customized Plot

- Create a plot similar to the picture using the given data
 - This is to show that marker colors, sizes and types can be given as arrays – the elements are applied sequentially

```
x = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]  
y = [3, 2, 5, 7, 4, 2, 4, 6, 7, 8]  
y_radius = [10, 20, 4, 15, 9, 9, 14, 8, 4, 9]
```

```
# Note that s (for size) represents the area, not radius  
plt.scatter(x, y, s = np.array(y_radius) ** 2, color = "red")  
plt.plot(x, y, linestyle = "dotted", color = "red")  
  
plt.xlim(np.min(x) - 1.3, np.max(x) + 1.3)  
plt.ylim(np.min(y) - 1.3, np.max(y) + 1.3)  
plt.xlabel("random x")  
plt.ylabel("random y")  
  
plt.show()
```



* Lab: Playing with matplotlib

- A very good part of matplotlib are the examples
 - See them [here](#)
- Many examples of common use cases
 - Creating multiple plots
 - Different types of plots: violin plot, residual plot, heatmap, etc.
 - Usages of color, shaded area, markers, labels, etc.
- Play with some of these examples to get a feel of what you can do with matplotlib
- Customize some of the examples
 - Read the docs to see all parameters



Exploratory Data Analysis (EDA)

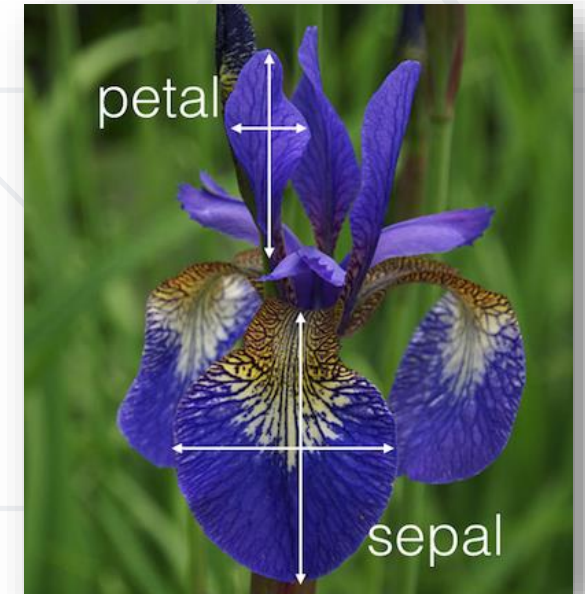
Making Sense of our Data

- A process to see what the data can tell us
 - Not tied to formal data modelling or hypothesis testing
- Many people have written about this
 - Most notably, John Tukey (1961)
- Like data cleaning, relies heavily on the scientist's intuition
- Objectives
 - Suggest hypotheses
 - Assess assumptions on which models will be based
 - Aid selection of features (feature engineering)
 - Provide a basis for further data collections

- Good for explaining the dataset visually
 - Show distributions, relations, comparisons and causality even in multivariate data
- Principles of analytic (scientific) graphs
 - Show **comparisons**
 - Show **causality**
 - Correlation does not imply causation
 - Show **multivariate data** (many variables)
 - Integrate evidence from **multiple sources**
 - Describe and **document** evidence
 - **"Content is king"**
 - We have to have something interesting to report
- These principles apply to EDA as well

Exercise: Exploration of the Iris Dataset

- One of the most famous datasets in data science
 - Sizes of petals and sepals (see picture) for three classes of iris flowers
- Read, inspect and clean the *dataset*
 - Using the data cleaning approaches you already know
 - **Can we predict classes from sizes?**
- Inspect the distributions
 - Plot histograms and boxplots, print stats
 - Try different plot settings
 - Compare the quantities – scatterplots
 - In some cases, a "brute force" method might be useful – compare everything against everything else
 - Plot a correlation matrix



Exercise: Exploration of the Iris Dataset

- Usually, we first perform univariate analysis, then go on to find correlations
 - Plot the entire distribution first
 - Start to break down by factors (in this case – the iris types)
 - Create additional columns if needed (data transforms)
 - Apply grouping, averaging and summing over groups to get an idea of possible "clusters" in the data
 - Inspect and plot certain data ranges (using filtering)
 - **Have fun with the data but don't forget the original question!**
- Next steps
 - After exploratory data analysis, we're usually able to form a hypothesis (a pair of hypotheses), model the data and check against the hypothesis
 - In other cases we can produce different visuals, graphics and dashboards to be used by others

* Exercise: Exploration of the Iris Dataset

- We can also plot beautiful graphics using other packages (not matplotlib)
- An example of one such package is seaborn
 - Contains utility functions for some commonly used plots
 - Based on matplotlib
 - Read the docs [here](#)
 - It shows how to plot different distributions on their own and together, and also includes a little tutorial on an algorithm called KDE (kernel density estimation)
 - It also has other [tutorials](#) (such as [plotting linear correlations](#))
 - It produces good-looking graphics but can lack customizability in some cases
- Other examples: [bokeh](#), [plot.ly](#), [ggplot](#)
 - And many more

Summary

- Main Concepts and Rules
- Creating Simple Plots
- Real-Life Examples: Good and Bad
- Customizing Plots
- Exploratory Data Analysis
 - Basic Guidelines
 - EDA as Part of the Data Science Process



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